Lab 4 – System modeling

Objectives: Understanding System modeling

Tasks:

- Learn how to use a UML tool
- Draw use cases in your project → Document
- Design ERD → Document
- Draw class diagrams → Document
- Submit your lab report

UML Tools

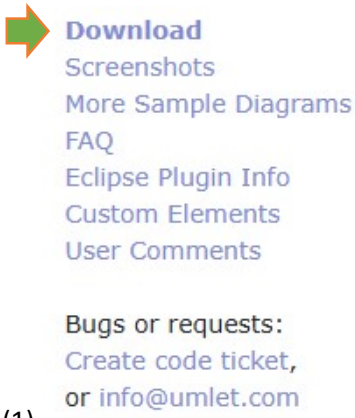
- https://www.visual-paradigm.com/features/
- https://www.umlet.com
- https://creately.com/app/

Tutorials

Learn how to use a UML tool

How to start UMLet?

Download the UMLet file from http://www.umlet.com/



(1)

(2)

More…

- Video tutorial: Basic Use and Custom Elements (https://www.youtube.com/watch?v=3UHZedDtr28)
- Custom UML elements: http://www.umlet.com/ce/ce.htm
- Online sample diagrams: http://www.itmeyer.at/umlet/uml2/

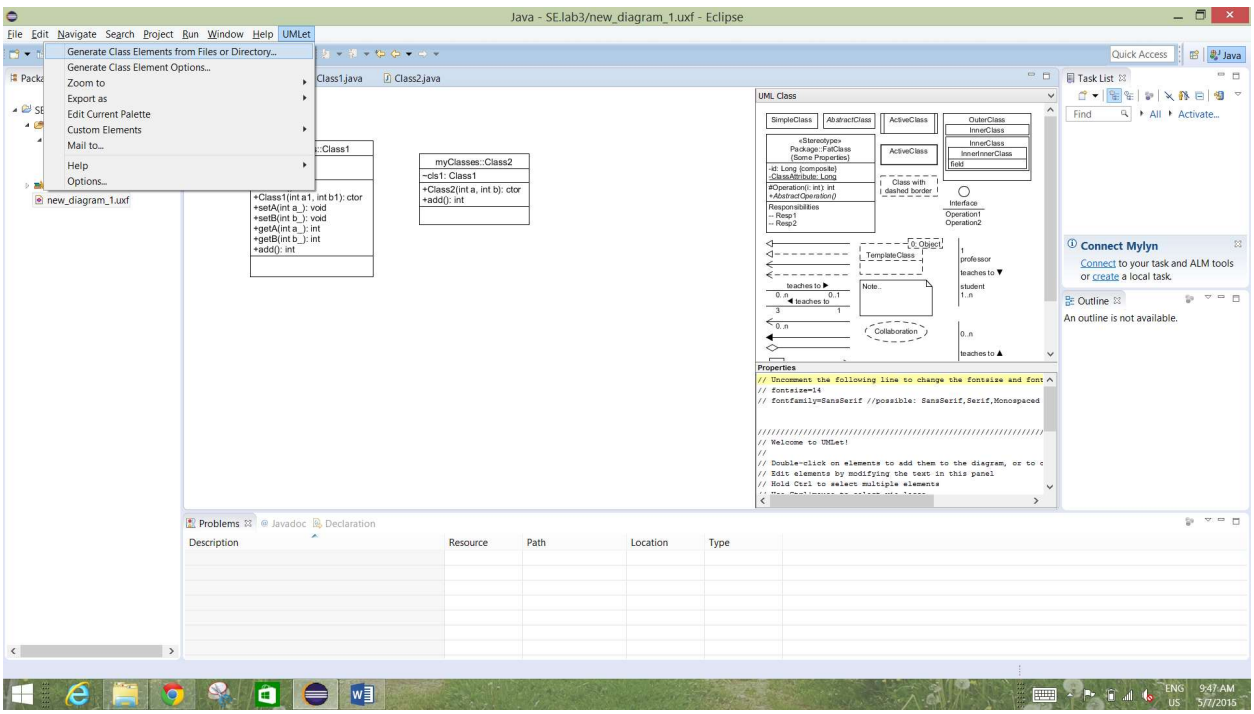
How to create use cases?

Referring to…

- UMLet tutorial for Context, UC, and Sequence Diagrams: https://www.youtube.com/watch?v=HVSxE296JLc
- Ex: https://www.itmeyer.at/umlet/uml2/UseCaseDiagram.html

How to create class diagrams?

- Generate Class Elements from Files or Directory…



- Choose available java class files to generate class elements.
- An example of using the UMLet plugin for Eclipse to make UML class diagrams: https://www.youtube.com/watch?v=7LDiw9K_KKo

More References on UMLet

[1] Examples of different types of UML diagrams using UMLet: http://www.itmeyer.at/umlet/uml2

[2] Sample diagrams in UMLet: http://www.itmeyer.at/umlet/

Use Case Diagram

Use cases describe the interaction between the system and external users that leads to achieving particular goals.

The use case diagram includes the following main elements:

- **Use cases**: Usually drawn with ovals, use cases represent different use scenarios that actors might have with the system (log in, make a purchase, view items, etc.)
- **System boundaries**: Boundaries are outlined by the box that groups various use cases in a system.
- **Actors**: These figures depict external users (people or systems) interacting with the system.
- **Relationships** are drawn with lines showing different types of relationships between actors and use cases.

Tutorials:

- Tutorial about Use Case Diagram:
  - https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-use-case-diagram/
  - https://creately.com/blog/diagrams/use-case-diagram-tutorial/
- How to draw Use Case Diagram using Visual Paradigm: https://www.visual-paradigm.com/support/documents/vpuserguide/94/2575/6362_drawinguseca.html
- Some examples: https://circle.visual-paradigm.com/category/use-case/

Entity Relationship Diagram

- Tutorials about ERD:
  - https://www.visual-paradigm.com/guide/data-modeling/what-is-entity-relationship-diagram/
  - https://creately.com/blog/diagrams/er-diagrams-tutorial/
- How to draw ERD using the Visual Paradigm tool: https://www.visual-paradigm.com/tutorials/how-to-model-relational-database-with-erd.jsp
- Some examples: https://circle.visual-paradigm.com/category/entity-relationship-diagram/

Class Diagram

Class diagrams are used when developing an object-oriented system model to show the classes in a system and the associations between these classes. An association is a link between classes that indicates a relationship between these classes.

- Tutorials about Class Diagram:
  - https://www.visual-paradigm.com/guide/uml-unified-modeling-language/uml-class-diagram-tutorial/
  - https://creately.com/blog/diagrams/class-diagram-tutorial/
- How to draw a class diagram using the Visual Paradigm tool: https://www.visual-paradigm.com/support/documents/vpuserguide/94/2576/7190_drawingclass.html
- Some examples: https://circle.visual-paradigm.com/category/class-diagram/